

**RAFFLES INSTITUTION**

**2014 Year 6 Preliminary Examination**  
Higher 1

CANDIDATE  
NAME

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CIVICS  
GROUP

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INDEX  
NUMBER

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**BIOLOGY**

Paper 2 Core Paper

**8875/02**

**15<sup>th</sup> SEPTEMBER 2014**

**2 hours**

Additional materials: Answer Paper

**READ THESE INSTRUCTIONS FIRST**

Write your index number, CT group & name on all the work you hand in.  
Write in dark blue or black pen on both sides of the paper.  
You may use a soft pencil for any diagrams, graphs or rough working.  
Do not use staples, paper clips, highlighters, glue or correction fluid.

**Section A**

Answer **all** questions.

**Section B**

Answer **ONE** question.

At the end of the examination, **hand in your essay SEPARATELY**.  
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |     |
|--------------------|-----|
| Section A          |     |
| 1                  | /8  |
| 2                  | /6  |
| 3                  | /9  |
| 4                  | /7  |
| 5                  | /10 |
| Section B          |     |
| 6 or 7             | /20 |
| Total              | /60 |

This document consists of **15** printed pages

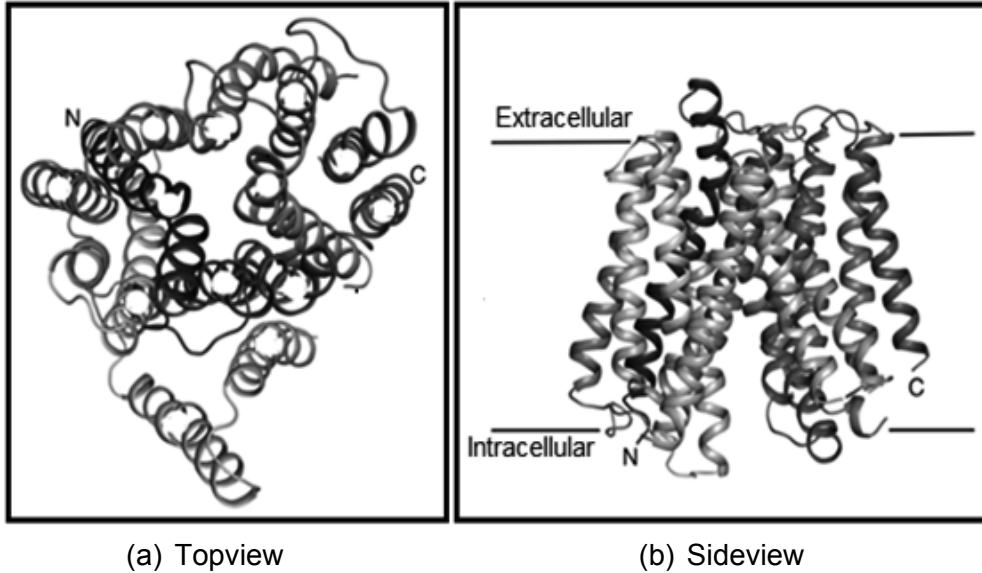


**Section A**

Answer **all** the questions in this section.

- 1** Differences in ion concentrations are maintained across cell membranes. These differences are many cellular processes.

Fig. 1.1(a) shows a top-view diagram of the membrane protein and Fig. 1.1(b) shows its relative position on the cell surface membrane.



**Fig. 1.1**

- (a)** With reference to Fig. 1.1 and your knowledge on membrane protein structure, describe one key structural features of this transport protein that allows it to perform its function.

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.....[1]

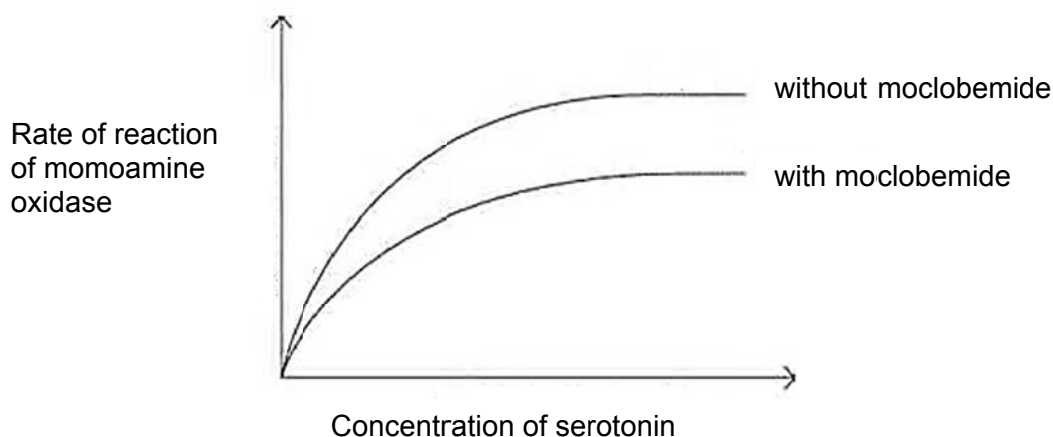
- (b)** In cells, the distribution of an ion is often unequal across the cell membrane. Describe how this unequal concentration of ion is maintained across the membrane.

.....

.....

.....[2]

- (c) Serotonin is a molecule that improves the mood of a person by stimulating the pleasure centre of the brain. Monoamine oxidase is an enzyme that breaks down serotonin. Moclobemide is an inhibitor of monoamine oxidase and was developed as an antidepressant. Fig. 1.2 below shows the effect of moclobemide on the rate of the enzyme-catalyzed reactions.



**Fig. 1.2**

- (i) Explain the effect of moclobemide on the rate of reaction of monoamine oxidase.

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.....[4]

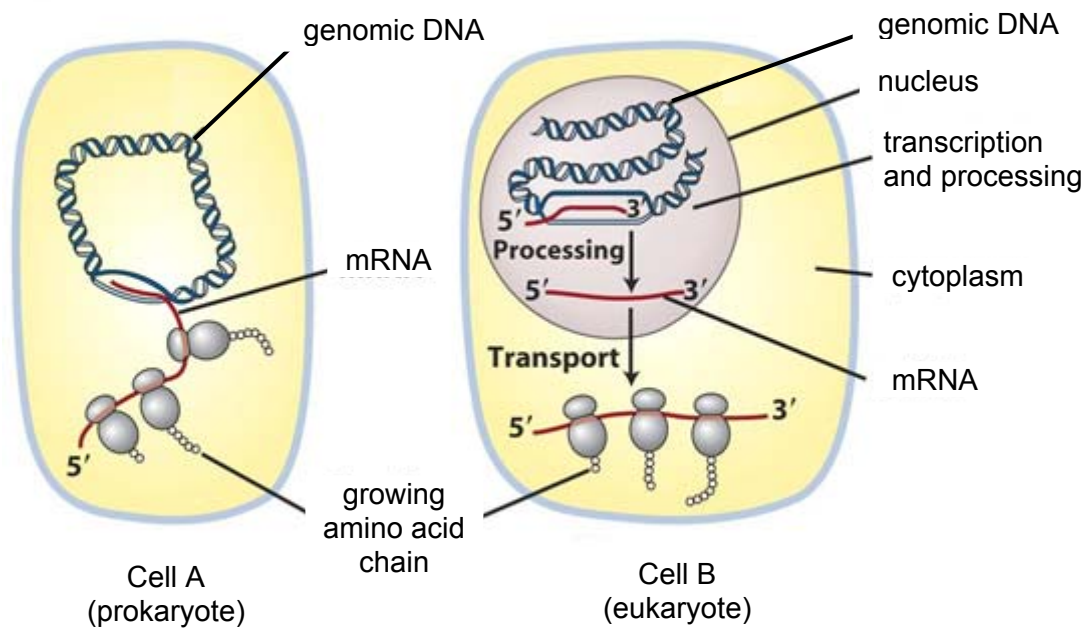
- (ii) Suggest how Moclobemide acts as an antidepressant.

.....

.....[1]

[Total : 8]

- 2 A student identified cell A as a prokaryotic cell as it does not contain a membrane-bound nucleus and cell B as a eukaryotic cell as it contains a membrane-bound nucleus.

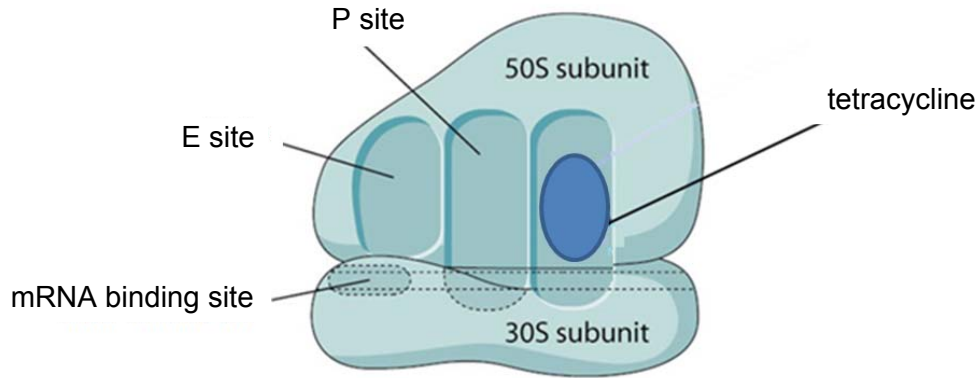


**Fig. 2.1**

- (a) With reference to the processes occurring within the cells shown in Fig. 2.1, give another reasons why cell A is likely to be a prokaryotic cell.

.....  
 .....[1]

- (b) Tetracycline is known to prevent bacterial growth by interfering with protein synthesis.



**Fig. 2.2**

- (i) With reference to Fig. 2.2, explain how tetracycline prevents protein synthesis.

.....

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.....[3]

- (ii) Suggest why tetracycline has no effect on human cells.

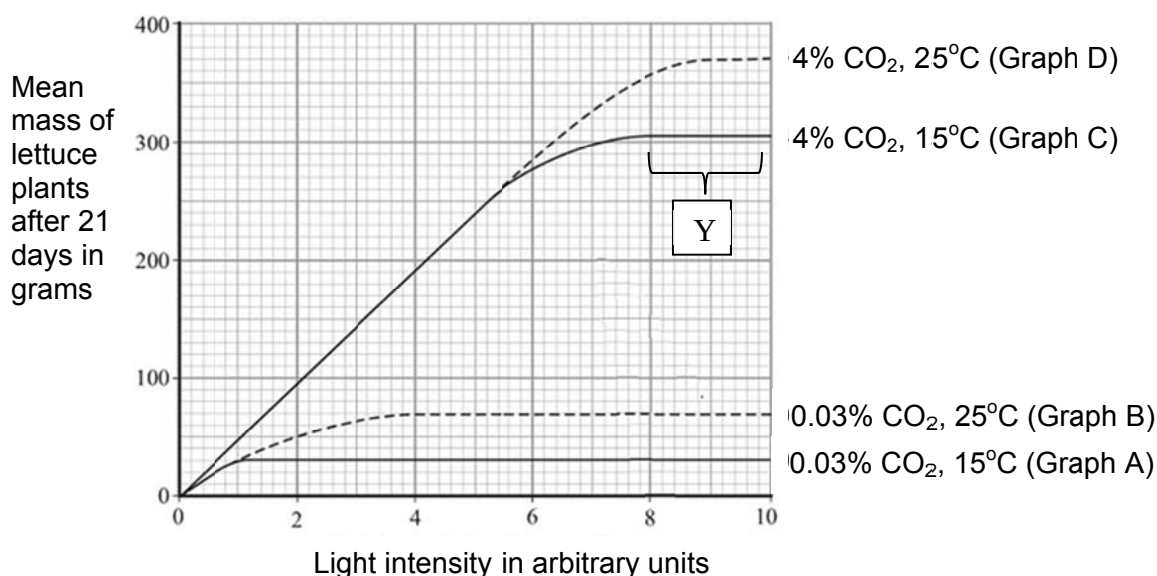
.....

.....[1]

- (iii) Furvina® is an antibiotic that inhibits initiation of translation. Indicate with an "X" on Fig. 2.2, the region where Furvina® binds. [1]

[Total : 6]

- 3 An experiment was conducted to investigate how various factors affect the rate of photosynthesis in lettuce. Fig. 3.2 below shows the results of the experiments conducted.



**Fig. 3.2**

- (a) (i) With reference to Fig. 2.2, state the best conditions for the growth of lettuce.  
.....[1]

- (ii) Explain the region marked Y.  
.....  
.....  
.....  
.....[2]

- (iii) Describe and explain the effect of increasing carbon dioxide concentration on the mean mass of lettuce after 21 days at 25°C.  
.....  
.....  
.....  
.....  
.....  
.....  
.....  
.....[4]

- (iv) The average carbon dioxide content of the atmosphere is 0.035%. Using this fact, and the information given in Fig. 3.2, what conclusion can be made about how carbon dioxide affects photosynthesis in the natural environment?

.....

.....

.....

.....[2]

[Total : 9]

- 4 Two pure bred varieties of a rice plant, Y and Z have been developed by the Raffles Rice Research Institute (R<sup>3</sup>I). Variety Y plants with heights ranging from 21-25 cm, have round leaves and are glyphosate resistant. Variety Z plants with heights ranging from 56-60cm, have wrinkled leaves and are glyphosate sensitive.

When variety Z was pollinated by variety Y, all its progeny are glyphosate resistant and round leaves. The height distribution of the F<sub>1</sub> generation are shown in the Table 6.1 below:

| height/cm | number of F <sub>1</sub> plants |
|-----------|---------------------------------|
| 26-30     | 1                               |
| 31-35     | 10                              |
| 36-40     | 54                              |
| 41-45     | 51                              |
| 46-50     | 8                               |
| 51-55     | 3                               |

**Table 6.1**

Table 6.2 shows the phenotypes of the F<sub>2</sub> progeny obtained when the F<sub>1</sub> plants were self-pollinated.

| height/cm        | number of F <sub>2</sub> progeny    |                                        |                                   |                                      |
|------------------|-------------------------------------|----------------------------------------|-----------------------------------|--------------------------------------|
|                  | round leaves & glyphosate resistant | wrinkled leaves & glyphosate resistant | round leaf & glyphosate sensitive | wrinkled leaf & glyphosate sensitive |
| 21-25            | 15                                  | 1                                      | 0                                 | 5                                    |
| 26-30            | 88                                  | 2                                      | 2                                 | 28                                   |
| 31-35            | 114                                 | 4                                      | 3                                 | 35                                   |
| 36-40            | 152                                 | 5                                      | 5                                 | 48                                   |
| 41-45            | 152                                 | 5                                      | 6                                 | 47                                   |
| 45-50            | 116                                 | 4                                      | 4                                 | 36                                   |
| 51-55            | 86                                  | 2                                      | 3                                 | 27                                   |
| 56-60            | 13                                  | 0                                      | 0                                 | 4                                    |
| <b>Sub-total</b> | <b>736</b>                          | <b>23</b>                              | <b>23</b>                         | <b>230</b>                           |

**Table 4.2**

- (a) State whether the alleles that code for round leaves and glyphosate resistance are dominant or recessive. Explain your answer.

.....

.....

.....

..... [2]

- (b) In cats, the gene controlling coat colour is sex-linked. The coat of male cats are either black or ginger while those of females are black, ginger or tortoiseshell (patchwork pattern of black, ginger and cream).

In a litter of kittens found in the basement of Raffles Rice Institute, half the number of females were tortoiseshell and the other half were black; half the number of males were black and the other half were ginger. Using a genetic diagram, show how mating between two parents of different genotype and phenotype, results in the phenotypic ratio as observed in this litter of kittens. [4]

- (d) It is often difficult to determine if coat colour is autosomal or sex-linked. Describe a way to determine if coat colour is autosomal or sex-linked.

.....

..... [1]

[Total : 7]

- 5 The Polymerase Chain Reaction (PCR) is used to amplify small amounts of DNA. Fig 5.1 shows the different stages of the PCR.

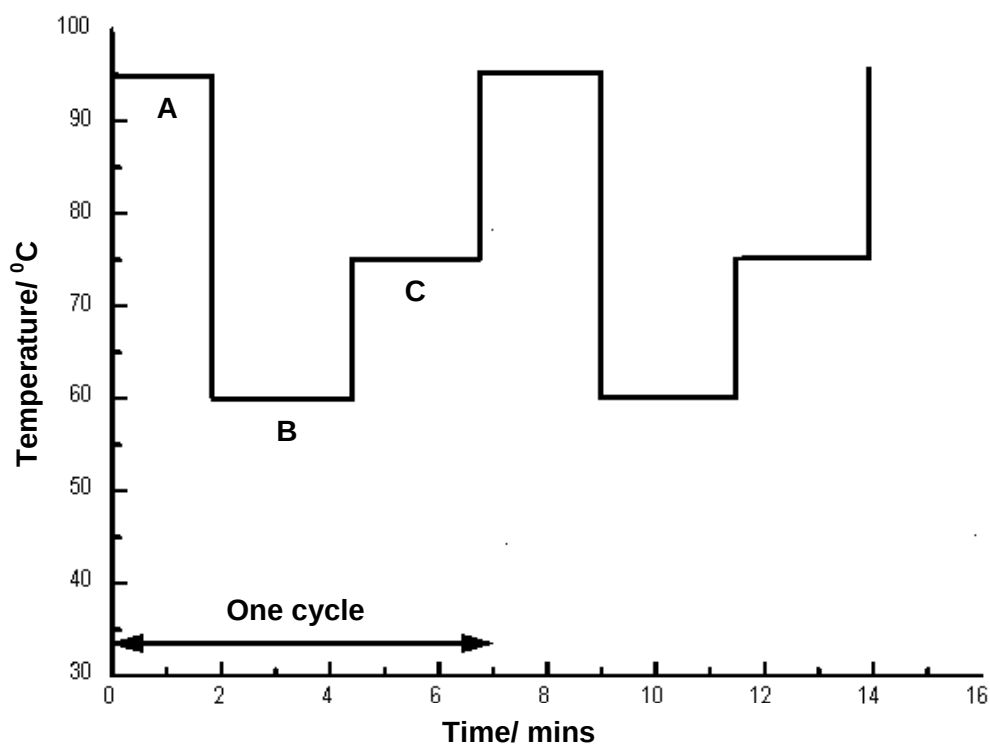


Fig. 5.1

- (a) With reference to Fig. 5.1, describe stage A.

.....  
 .....[2]

- (b) The following were added into the PCR mixture before being placed in a thermocycler:

- DNA template
- enzyme X
- dNTPs
- forward and reverse primers
- magnesium chloride + buffer solution

- (i) Name the stage of the PCR which would be affected if enzyme X was not added to the PCR master mixture.

.....[1]

- (ii) Identify enzyme X and explain why it is suitable for use in PCR.

.....  
 .....[1]

- (c) Plasmids are circular pieces of extra chromosomal DNA, found in many bacteria.

pUC19 is a plasmid that is frequently used for gene cloning. As shown in Fig. 5.2, the pUC19 plasmid has:

- an origin of replication (ori);
- a *lac Z* gene with its promoter. The *lac Z* gene codes for the  $\alpha$  peptide fragment of the enzyme  $\beta$ -galactosidase;
- a short length of DNA called the 'Multiple Cloning Site' (MCS) which contains the recognition sites of 3 different restriction enzymes;
- a gene conferring resistance to the antibiotic ampicillin ( $\text{amp}^R$ ).

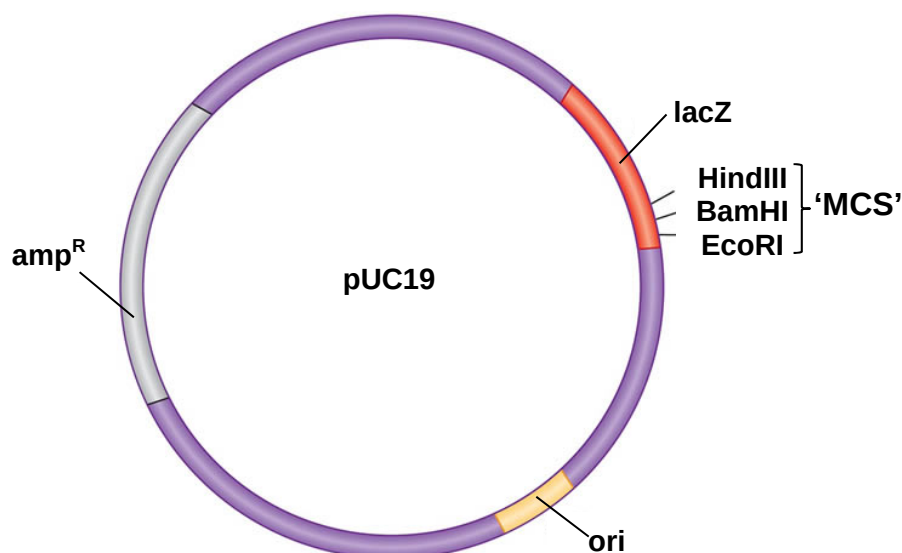


Fig. 5.2

- (i) Explain the purpose of the  $\text{amp}^R$  gene on pUC19.

.....  
 .....  
 .....[2]

Recombinant plasmids were created by cloning the gene coding for human insulin chain B into the pUC19 plasmid.

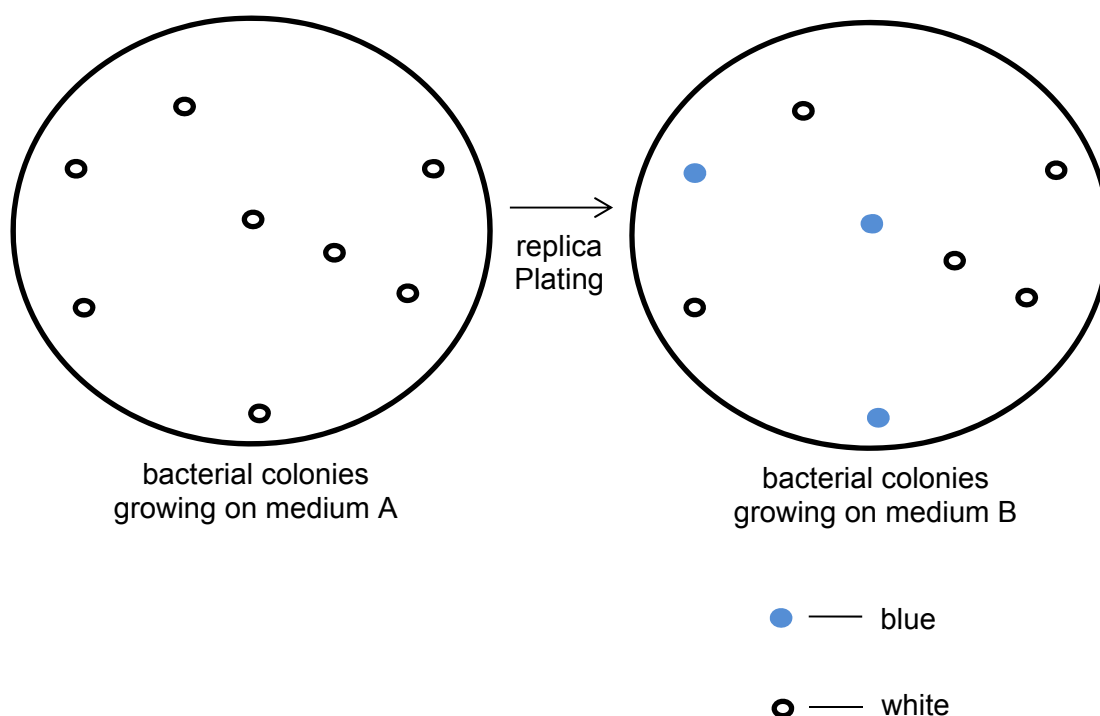
Table 5.1 shows the composition of two different nutrient media.

| medium      | A                        | B                                                       |
|-------------|--------------------------|---------------------------------------------------------|
| composition | nutrient agar+ampicillin | nutrient agar with lactose +<br>Compound S + ampicillin |

**Table 5.1**

Competent *Escherischa coli* (*E.coli*) were transformed with recombinant plasmids and grown on medium A.

Bacterial colonies growing on medium A were replica plated onto medium B. Fig 5.3 shows the growth of the resultant bacterial colonies on different media.



**Fig. 5.3**

(ii) With reference to Table 5.1 and Fig. 5.3, identify the compound S.

.....[1]

(iii) Explain why compound S was used in medium B.

.....

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.....[3]

[Total:10]

**Section B**  
**Answer EITHER 6 OR 7.**

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 6 (a)** Describe the significance of chromosome behavior during the various stages of mitosis. [6]
- (b)** Compare oxidative phosphorylation with photophosphorylation. [6]
- (c)** Explain how homology supports Darwin's theory of evolution (descent with modification)? [8]
- [Total: 20]

- 7 (a)** Discuss the significance of NAD and NADP. [8]
- (b)** Compare the differences between the structures of cellulose and insulin. [4]
- (c)** Outline the role of proteins in the movement of molecules in and out of a cell. [8]
- [Total: 20]

**End of Paper**